The Salish Sea Dreaming — Briefing for Prav

March 12, 2026 — from Darren

Hey Prav — sorry we couldn't connect today. Here's everything you need to get oriented and start testing. The most important thing you can do right now is **get Autolume running with the smoke test checkpoint** (Section 6 has the quickstart). That way when the real trained models land in the next few days, you're ready to go.

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1. What We're Building (April 10–26)

"Digital Ecologies: Bridging Nature and Technology" — Mahon Hall, Salt Spring. Opens **April 10**.

The Salish Sea Dreaming is an immersive AI art installation where the Salish Sea uses technology to perceive itself. Audiences enter a co-dreaming space — the ecosystem's consciousness made visible through generative imagery, presence-responsive visuals, and biosonification.

Three visual sources feed the generative system:

1. **Briony Penn's** ecological watercolors → visual DNA for a custom StyleGAN2 model (training now)
2. **David Denning's** photographic archive → "long-term bioregional witnessing" via LoRA+img2img style transfer
3. **Moonfish Media** underwater cinematography → herring, salmon, marine habitat footage as reference/projection

The live stack: Autolume (StyleGAN2 inference) → NDI → TouchDesigner + **Resolume Arena** (projection mapping/video mixing) → 1–2 projectors onto 3–6m wall.

Sound: You + TBC collaborator — Ableton Live + Max for Live biosonification.

Two modes: Autonomous Installation (runs unattended during gallery hours) + Live Performance (you performing real-time).

Spatial spec from your Tech Rider: 4m × 4m floor, 3–6m projection wall, 1–2 projectors, stereo audio, dim lighting.

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2. How the Training Pipeline Works

We're using **StyleGAN2** via Autolume from SFU MetaCreation Lab — a GAN that learns visual patterns from a dataset and generates new images in the same style.

```
539 marine photos          54 Briony watercolors          Live performance
(iNaturalist, CC)          (artist permission)            (Autolume + TD)
      |                          |                              |
      ▼                          ▼                              ▼
 Base training ────────→  Fine-tune (resume) ────────→  .pkl → Autolume → NDI → TD
 539 photos, H200          54 Briony paintings           RTX 3060 inference
 Currently running         Next step                     Real-time, audio-reactive
 → extending to 1000+      Same hardware                 OSC/MIDI control
```

Stage 1 — Base model (539 marine photos): Train from scratch on curated underwater/nearshore photos (37 Salish Sea species). Teaches the model what marine life *looks like* — colour palette, textures, forms, light. Output: `base-underwater-v1.pkl`.

Stage 2 — Fine-tune (54 Briony watercolors): Resume from base checkpoint on Briony's art. Shifts output from photographic → painterly. Multiple snapshots give a **gradient from photographic to painterly** — you pick the sweet spot or interpolate live.

Stage 3 — Inference (you in Autolume): Load the `.pkl` into Autolume. Real-time latent space navigation, audio-reactive, OSC/MIDI, NDI out to TouchDesigner. This is what the audience sees.

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3. Where We Are Right Now (March 12)

Base model training is running on the TELUS H200 GPU — 539-image marine photo corpus. Currently ~46% through the first 200 kimg target. Training is stable at 100% GPU utilization.

Arshia (SFU MetaCreation Lab researcher — he built Autolume) is advising us. His key guidance:

- Extend base training to **kimg=1000+** (current 200 is just a v1 checkpoint)
- **LoRA + img2img** may be better than GAN fine-tune for applying Briony's style to David Denning's photos
- 54 Briony images is low for fine-tuning — experimental, but worth trying
- Mac Autolume version is **in development** but not ready yet — Windows/Linux only for now
- He has access to **Alliance Canada's 4x H100 cluster** as backup to TELUS

Timeline for you getting real models:

- v1 base checkpoint (200 kimg): ~12-24 hours from now
- Extended base (1000+ kimg): ~3-4 days
- Briony fine-tune: starts after base, ~1 hour per run, multiple snapshots

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4. The GitHub Repo

github.com/DarrenZal/salish-sea-dreaming

What	Link
Full repo	github.com/DarrenZal/salish-sea-dreaming
Training data + provenance	training-data/
Corpus specs + smoke test results	training-data/README.md
Full technical architecture	docs/autolume-integration.md
Briony art archive (git-lfs)	VisualArt/Brionny/
Scripts + tools	scripts/ · tools/

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5. The Training Data

Briony fine-tune corpus — 54 watercolors at 512×512

- Browse all 54 PNGs on GitHub
- provenance.csv — tracks source painting, crop method, artist permission
- Expanded from marine-only → all ecological watercolors (salmon-forest, camas, landscapes)

Marine photo base corpus — 539 photos at 512×512

- marine-photo-base.zip on Google Drive — 64 MB download
- QC'd from 740 iNaturalist candidates across 37 Salish Sea species
- rejects.csv — 201 rejected images with reasons

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6. YOUR FIRST STEP: Get the Smoke Test PKL Running in Autolume

This is the most useful thing you can do right now. We have a smoke test checkpoint (undertrained, but it loads and runs). Getting Autolume working with this means when the real models land, you just swap the file.

Quickstart Guide

Full step-by-step: **docs/autolume-quickstart.md on GitHub**

TL;DR:

1. Install CUDA 12.8 + C++ Build Tools + Miniconda
2. Clone Autolume: `git clone https://github.com/Metacreation-Lab/autolume.git`
3. Download checkpoint: **network-snapshot-000025.pkl** (347 MB) → put in `models/`
4. Run: `python main.py`

Available Checkpoints (smoke test — not final)

Checkpoint	Download	Notes
network-snapshot-000000.pkl	347 MB	Initial (random noise)
network-snapshot-000020.pkl	347 MB	20 kimg
network-snapshot-000025.pkl	**347 MB**	**25 kimg — start with this**

Dataset zips (also on Drive)

File	Download
briony-marine-colour.zip (54 watercolors)	27 MB
marine-photo-base.zip (539 photos)	64 MB

SSD Shared Drive folder — all assets live here

Compatibility Warning

Our PKL was trained with the NVlabs StyleGAN3 codebase. Autolume uses its own fork. If the checkpoint **fails to load**, let me know immediately — we'd need to retrain with Autolume's own training pipeline. Better to find this out now with the smoke test than after days of training.

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7. Installation Hardware

Option	Pros	Cons

Your RTX 3060 laptop	Works now, can test today	Not ideal for 24/7 exhibition
Dedicated desktop (RTX 3060/3070)	Reliable, affordable used	Need to source + budget
Mac Mini M4 (MOVE37XR inventory)	Already available	**Won't work** — Autolume is Windows/Linux only (Mac version in dev, not ready)

Key constraint: Autolume requires **NVIDIA GPU with CUDA**. No Apple Silicon support yet.

You mentioned grant budget may be available. Even a used desktop with an RTX 3060 would be solid for continuous exhibition mode.

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8. Scope Alignment — Your Exhibition Outline

I reviewed your three project docs (Collaboration Outline, Exhibition Outline, Tech Rider). Really well articulated. A few things I want to make sure we're aligned on for April:

Explicitly excluded for April (from your Exhibition Outline):

- No networked GPUs or cloud inference
- No live ecological data feeds
- No multi-room installation
- No AI agents or chat interfaces

This aligns well with where we are. The April version is: pre-trained models running locally on an NVIDIA machine, Autolume inference, TD + Resolume compositing, Kinect presence, Ableton sound. Clean and focused.

"Integrity over ambition" / "succeeds by doing less, not more" — agreed. Let's nail this.

Things your docs mention that aren't in our technical architecture yet:

- Resolume Arena (I've added it to our docs now)
- Sound layer details (Ableton + Max for Live biosonification)
- Two operational modes (autonomous vs live performance)

Things our side has that aren't in your docs yet:

- TELUS H200 training progress (base model running)
- Arshia's involvement + MetaCreation Lab advisory
- LoRA + img2img as alternative to pure GAN fine-tune
- Alliance Canada backup GPU access

We should do a joint pass to sync these up.

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9. Phase 2 Venues (from your docs)

Venue	Date
Indigenomics Impact	May 2026
MOVE37XR Symposium	October 2026

DEV CON ETH Mumbai	November 2026

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10. What You Can Do Right Now

Priority 1 — Test Autolume:

1. Download 000025.pkl (347 MB)
2. Follow the quickstart guide
3. Tell me if it loads or fails — this is critical info

Priority 2 — Hardware decision:

- What machine for the April 10 exhibition? Your laptop or dedicated desktop?
- Is grant budget available?

Priority 3 — Review the repo:

- Browse the GitHub repo
- Look at the 54 Briony training images
- Read the full architecture doc

What I'm Doing Next

- Monitoring base training on TELUS (should finish 200 kimg overnight)
- Download v1 checkpoint and share with you on Drive
- Resume training to kimg=1000+ (Arshia's recommendation)
- Start Briony fine-tune from base checkpoint — will share snapshots as they come
- Friday 12:30 call with Arshia — review training progress, discuss LoRA approach
- Friday TELUS meeting — push for environment improvements

I'll share new checkpoints on the Drive folder as they land.

Questions? Hit me on Signal anytime.